

ABRA — Project Summary

Version 1.0.0 · 2026-09-24 · Will Hooper Line: abra/regmc — CHANGELOG-REGMC.md .

1.0.0 — MEDICHAM IS CERTIFIED ON REG M-C. THE ENGINE IS THE FOUNDATION; THE SEARCH IS THE POINT. THE SOLVER NOW SEARCHES ON IT, AND THE FINISH LINE IS HIGH ON THE LADDER.

ABRA builds a player for Pokémon Champions VGC, Reg M-C, open team sheets, and takes it up the Showdown ladder (gen9championsvgc2026regmcho3 , account medicham32 , cleared with Showdown staff). The plan runs in three steps:

- MEDICHAM is correct.** ABRA's simulator agrees with Showdown on every instrument that measures it on Reg M-C, and the gate is OPEN (§2).
- So the solver searches on it, live, every turn.** At team preview **CHOMP** solves the bring and the lead. Each turn **XATU** holds the belief over hidden information, **MAG and DODUO** narrow the candidate joint actions, **MILTANK** fills a payoff matrix by playouts on MEDICHAM and **SLOWKING** solves it; **HYPNO** is a capped exploit dial against human habits. **PORYGON2** is the value net, **MEW and MACHAMP** train it by self-play, and **ROTOM** is the live client.
- It climbs the ladder**, judged by a settled rating $\pm$ SD over many series and the per-series residual, never the peak.

both open sheets → CHOMP → bring 4, lead 2
each turn: XATU (worlds) → MEDICHAM legalActions → DODUO (prune) → MILTANK (playouts)
→ SLOWKING (equilibrium) → HYPNO (exploit dial) → sample → ROTOM → ladder

The plan, with the model registry and milestones, is solver/PLAN.md ; what landed is solver/LOG.md . **Reg M-B is retired** (Will, 2026-09-24). Its record is CHANGELOG.md , closed at 7.0.0, and the Reg M-B edition of this document is git show 1be7343c:docs/SUMMARY.md . Nothing here is comparable with a figure in it.

1. Every component, and the state it is in

component	what it is	state on 2026-09-24
MEDICHAM	ABRA's doubles simulator, and its solver API engine/medicham_api.js (clone , legalActions , step , terminal check, lean playouts)	Reg M-C GATE OPEN , all ten clauses (§2)
CHOMP	team-preview solver: both open sheets → a mixed strategy over the bring/lead options	to build (milestone M4). Rebuilt inside ABRA; the old ../CHOMP repository is reference only
XATU	belief over the opponent's back two and stat spreads	v1 built , store-only (§3)
MAG	per-slot action scorer — the human policy prior	v1 built , store-only (§3)
DODUO	joint coordinator — scores both slots as one joint action	v1 built , store-only (§3)
MILTANK	search harness: candidates, playouts on MEDICHAM, the payoff matrix, the clock	v1 built with a worker pool and lean playouts. Strength figures PRE-GATE, withheld (§3)
SLOWKING	per-turn simultaneous-move solver (regret matching and an exact LP)	v1 built , unit-tested (§3)
PORYGON2	value net: state → P(win)	to build (M5)
GARY / HYPNO	human habits by situation / the capped exploit dial	to build (M7)

component	what it is	state on 2026-09-24
MEW / MACHAMP	self-play factory / training loop	to build (M5)
WOBBUFFET / DUSK	exploitability best-responder / endgame tables	to build (M6)
GURU	meta analysis over the store	vo built , descriptive, store-only
DITTO	team builder	to build (M8)
ROTOM	the live ladder client; replaces <code>engine/mag_bot.js</code>	to build (M3)
ALAKAZAM / KADABRA	the assembled agent / the coach	to build
The store and the ingest	every Reg M-C game kept raw, collected hourly	LIVE (OPS)
The honesty machinery	<code>status.js</code> , <code>provenance.js</code> , <code>quarantine.js</code> , the documentation gate	LIVE
ABRA WORLD — the site	renders what the artifacts say	PAUSED

Store-only means the model never runs the simulator, so it never waited on the gate. Everything that plays games on MEDICHAM did wait, and anything it measured before the gate opened is withheld.

2. MEDICHAM on Reg M-C — what the gate read

All on engine release `eea5becc54eb` , census pin `123aa264f88d` , pool `data/team-pool-frozen-regmc` , authority the `pokemon-showdown-mc` checkout (CHANGELOG-REGMC 1.0.0; docs/_reports/2026-09-24-regmc-gate-final.md). `--games` is part of the sample definition, so each lattice is named by it.

instrument	reading (1.0.0)	read from
whole game, board-material, <code>--games</code> 1200 / 1600 / 1900	0/955, 0/1266, 0/1497 games part a board	<code>data/game-differential-regmc.json</code> , <code>data/game-differential.g1600-regmc.json</code> , <code>data/game-differential.g1900-regmc.json</code>
whole game, undeclared narration	0/955, 0/1266, 0/1497 ; the narration bar is stamped at zero	the same three, and <code>data/whole-game-baseline-regmc.json</code>
the damage differential	0/6000 at the midpoint, top, bottom and all 14 interior roll indices	<code>data/engine-diff-regmc.json</code>
deliberate roster	items 166/166 , abilities 210/214 , moves 510/511	<code>data/roster.items-regmc.json</code> , <code>data/roster.abilities-regmc.json</code> , <code>data/roster.moves-regmc.json</code>
every in-scope mechanic staged	0 diverge in 4,867 games, 0 threw	<code>data/all-mechanics-fire-regmc.json</code>
coverage	all 269 moves above 25 clicks measured (CHANGELOG-REGMC 1.0.0)	<code>node engine/quarantine.js --regulation regmc</code>
the gate	OPEN , 10 of 10 clauses (CHANGELOG-REGMC 1.0.0)	<code>node engine/quarantine.js --regulation regmc</code>

The four ability rows not counted as tested at 1.0.0 are declared, not hidden: three pass as announcement-only on recorded receipts, and one (Illusion) is deferred by the owner.

3. The solver — what is built and what it measured

Store-only models, measured on held-out players. These train on the Reg M-C human dataset (bo3, open sheets) with no simulator in their path, so their figures stand.

- **MAG v1 and DODUO v1** (CHANGELOG-REGMC 0.113.0). On the held-out test players' 25,477 exact joint actions, DODUO's joint log-loss is 2.730 (95% CI 2.699–2.761) against the vo prior's 2.921. The node forward pass matches Python to 2.1e-14; `solver/tests/test-mag-doduo.js` passes 3,826 of 3,826.
- **XATU v1** (CHANGELOG-REGMC 0.114.0). On 10,942 held-out sides, turn-1 log-loss on the true back pair is 1.348 against 1.792 for a uniform guess, and the true bring is never ruled out.

The recall figures behind those two lines are read from the tracked metrics file; the readout, with the command, is in `docs/MODELS.md` under *MAG v1 and DODUO v1*.

Models that play on MEDICHAM. SLOWKING v1, MILTANK v1 and the offline arena are built and unit-tested (CHANGELOG-REGMC 0.115.0): `test-slowking` 1,559/1,559, `test-miltank` 3,414/3,414, `test-arena` 15/15. **Every arena strength figure so far is PRE-GATE — it was played before release eaa5becc54eb — and is withheld, not captioned.** The re-run on that release is the first thing owed. No direction may be read into the absence.

4. The road to the ladder

The milestones are `solver/PLAN.md` §3, in dependency order. Where they stand:

milestone	what lands	state
Mo	Reg M-C gate open; solver API merged	gate OPEN (1.0.0); API merged; the mid-turn-choice callback still owed
M1	DODUO split out of MAG, XATU's bring posterior, GARY vo, MAG intersected with <code>legalActions</code>	MAG, DODUO and XATU v1 built; GARY and the <code>legalActions</code> intersection owed
M2	SLOWKING + MILTANK, one-ply matrix, rollout leaf	built; the post-gate SPRT against greedy one-turn and against MAG with no search is owed
M3	the clock and ROTOM on a local Showdown server	to build
M4	CHOMP vo, ALAKAZAM v1, the first ladder burn-in	to build; a ladder launch needs Will's OK
M5–M8	PORYGON2 and self-play, per-world opponent tables, HYPNO + GARY on the ladder, CHOMP v1, DITTO	to build

How it will be judged (`solver/PLAN.md` §5): offline SPRTs at equal wall-clock against each model's named baseline, then the per-series ladder residual $S - E$, read by SPRT, with the arm drawn per series by a seeded coin. The headline is the mean rating over the last N series $\pm$ SD. Never the peak.

5. The honest column

- **No strength claim is made.** The only strength figures that exist are PRE-GATE and withheld.
- **A zero on the gate is a statement about what was measured** — three lattices, a damage battery and a staged lab, on one release and one frozen pool. It is not a proof of equivalence.
- **Open team sheets only.** Closed sheets and bo1 are out of scope; nothing here says how the player does when the sheet is hidden.
- **Illusion is the one declared exclusion.**
- **XATU's spread narrowing is weak.** Two unknown spreads cannot be pinned from a replay; live play, where our own spread is known, should narrow more and has not been measured.

- **The ladder can resolve only large differences** — rating noise alone moves an account by tens of points, and a small improvement needs hundreds of series per arm to show (`solver/PLAN.md §5`, derived in `docs/_reports/2026-09-23-solver-research-humans-and-ladder.md`).
 - **The 1.0.0 release is drafted and held for Will to read.** It is on `draft/regmc-1.0.0` , not main.
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6. Why this is a MAJOR

A MAJOR here means the BASIS moved — the question the numbers answer. Two things a reader can no longer be told:

1. **That MEDICHAM is not correct on Reg M-C.** It is, on every instrument, so the withholding that rested on that sentence ends for the Reg M-C line at once: every Reg M-C artifact downstream of MEDICHAM becomes RE-RUNNABLE, not current, and nothing measured before `eea5becc54eb` may be quoted until it is re-run.
 2. **That ABRA is a meta-analysis platform whose preview tool lives in another repository.** The project is now a player: every model except MEDICHAM is rebuilt from scratch under `solver/` for open-sheet Reg M-C, and CHOMP is its preview solver. The Reg M-B models are retired, not repaired, and no figure of theirs links to one here.
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7. Where the current state is read

State is printed, never typed. Nothing in this document outranks what these print today.

```
node engine/status.js                # every figure, with the artifact it came from
node engine/quarantine.js --regulation regmc  # the Reg M-C gate, clause by clause
node engine/open_work.js              # every unclosed register row and measured defect
node engine/docs_scan.js --owed        # what the documents still owe the next major
```

`docs/RUNNING-NOTES.md` is the log between majors. The division ledgers —
`docs/{ENGINE,MEASURE,SOLVER,OPS,WEB}.md` — carry the working detail, `solver/LOG.md` the solver's narrative,
and `docs/_reports/` the dated accounts. None of them is state.